
A Project On
---Fighting Car--



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Introduction: Buckle up! Unleash the thrill of head-to-head combat with RC fighting cars! These aren't your average remote-controlled racers. We're talking about tricked-out vehicles built for battle, equipped with features like:

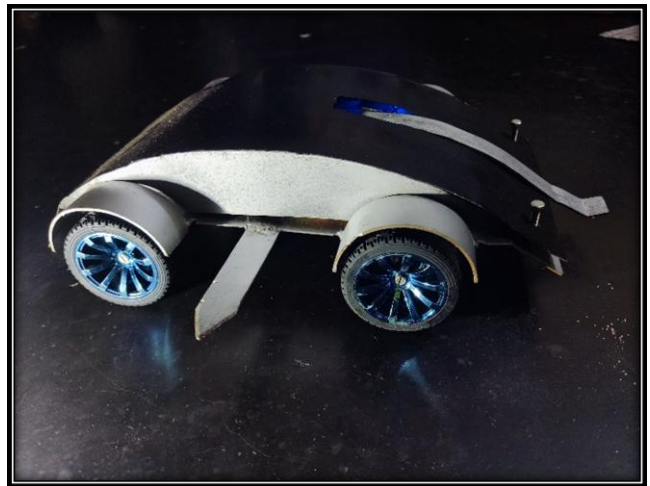
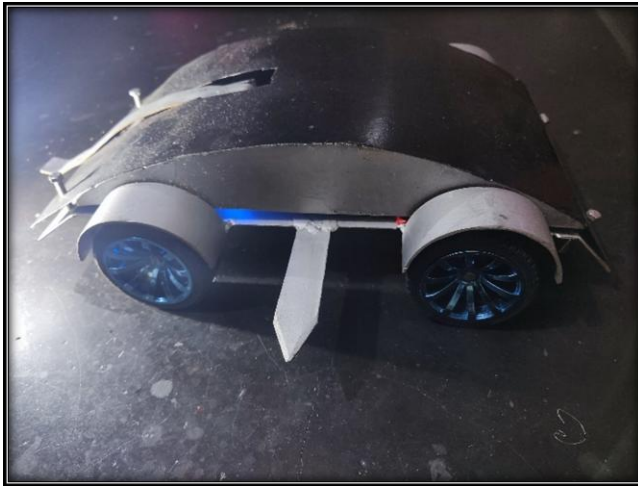
- **Durable builds:** Built to withstand crashes and collisions, these RC cars can take a beating and keep on battling.
- **Agile handling:** Perform strategic manoeuvres, dodge attacks, and outmanoeuvre your opponent with lightning-fast reflexes.

RC fighting cars are perfect for:

- **Sparking adrenaline-pumping competitions:** Challenge your friends and family to an epic RC duel!
- **Testing your skills:** Develop strategic thinking, quick reflexes, and master car control in the heat of battle.

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- **Supercharged fun:** Experience the joy of competition and friendly rivalry with a unique RC twist.

So, are you ready to enter the arena? Get your hands on an RC fighting car and prepare for battle!



Process Model:

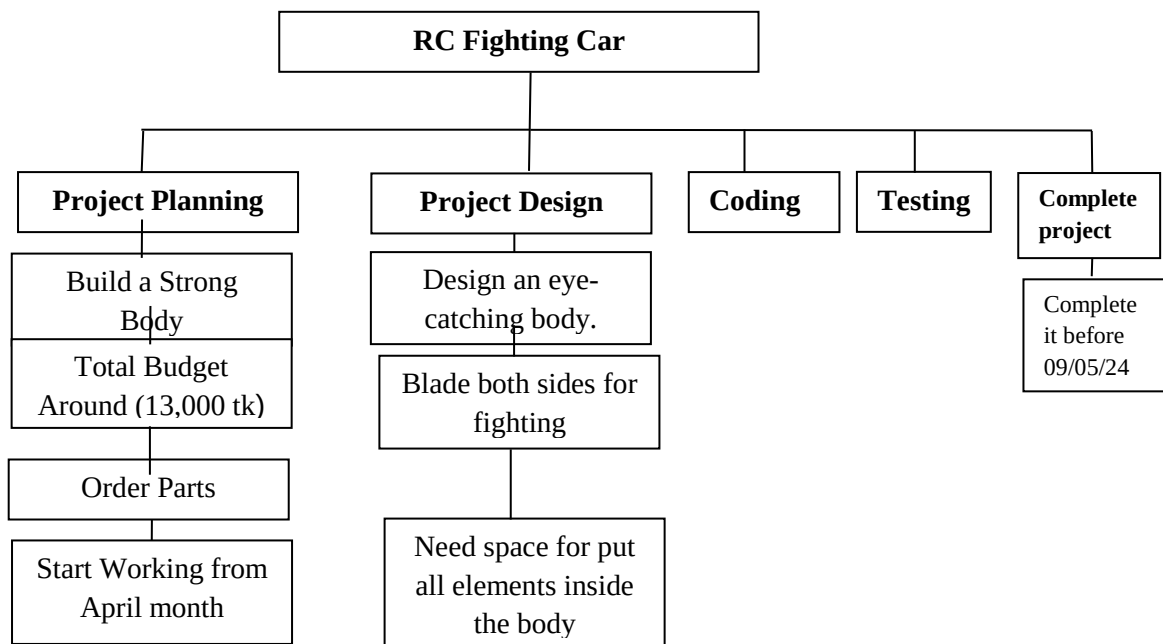
For Our project, we choose **The Linear/waterfall model** Because this is beneficial for us and for our project.

Here is the benefit of this model.

1. **Clear Structure:** The waterfall Methodology. flows a rigid structure with distinct phases, each having specific goals and deliverables. This sequential approach ensures a well-defined project path.
2. **Easy information transfer:** Because each stage in the waterfall model has a specific deliverable, passing information from one page to the next is straightforward. These clarity aids effective communication within the team.
3. **Stability:** The project scope remaining relatively static. allowing early determination of cost and timelines. changes to systems are minimal, keeping the cost of alternations low.
4. **Early goal setting:** Press is requirements definition enabled accurate Estimation of the project's total cost and timeline.
5. **Legacy systems:** Where stability and minimal changes are critical, the waterfront model ensure a systematic update process. It Allows for careful planning and execution without disruption, existing functionality.

In summary, the waterfall model provides structure, stability and clear communication, making it beneficial for our projects.

List of tasks:



- **Resorts and schedule planning:** Allocate resources. Develop a project schedule with milestone and deadlines.
- **Requirement and detail design:** Gather detailed requirement for the hardware. Create design specifics based on their requirements.
- **Engineer solution:** Develop the harder solution. Prototype and test the solution.
- **Order Hardware:** Procure necessary hardware components.
- **Hardware installation and configuration:** Install harder components. Configure setting and connections.
- **Application development and testing:** software, develop and test that interact with the hardware. Ensure capability and functionality.

Estimation for each task:

Start with an overall time estimate for the entire project.

First, we start discussing which part we need. Then we listed the part.

1. **Aminul Hoque Shawon** successfully finalized the structural blueprints that guided our entire workflow. Once his design phase was completed, the team transitioned to component sourcing. Furthermore, we secured a dedicated order for the body augmentation structures. Shawon's careful planning during the design stage ensured that all locally and digitally sourced materials fit together seamlessly for the build phase.

We have placed an order for body augmentation and buy some parts from the market what we needed. What we don't get the market, we ordered them on online shop.



2. **Anika Tahasin** successfully programmed, coded, and implemented the entire software infrastructure. Following deployment, she conducted rigorous testing and system observation to isolate technical errors and bugs. Once these issues were identified, the entire team collaborated to execute effective troubleshooting steps. She ultimately concluded her contributions by serving as the primary speaker, successfully demonstrating our work at the Project-Based Learning (PBL) Show.



3. **Anttu Majumder** is an assembler for our project. He and others member help to

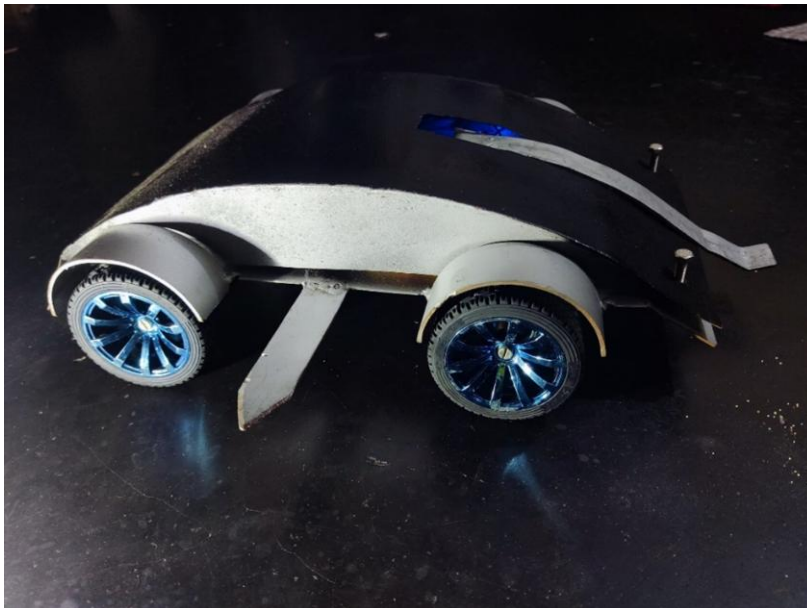


assemble all parts together.



4. **Tahmed Talukder Hemal** is a financial advisor for our project. She advice's us for minimal our project expenses and best product for minimal price.

After all member's hard work, we finally made our project.



Schedule the tasks:

Id	Task Name	Duration	Start	Finish
1	Planning			
2	Recruitment analysis	3 Days	Wed 03/04/24	Fri 05/04/24
3	Design	3 days	Sun 07/04/24	Wed 10/04/24
4	Body order & receive	4 Days	Thu 11/04/24	Mon 15/04/24
5	Component's order & receive	26 days	Fri 12/04/24	Sun 8/05/24
6	Assembling			
7	Battery assembling	1 Day	Wed 17/04/24	Wed 17/04/24
8	Battery connected BMS and charging	3 Days	Thu 18/04/24	Sun 21/04/24
9	Motor assembles	4 Days	Tue 23/04/24	Sat 27/04/24
10	All components assemble to the body	5 Days	Sun 28/04/24	Fri 03/05/24
11	Testing and solving			
12	Taste and solve the project problem	6 Days	Fri 03/05/24	Thu 09/05/24

List of milestones:

1. Concept Design:

- Finalize the conceptual design of the RC fighting car.

2. Mechanical Design Completion:

- Complete detailed mechanical design, including chassis, body and components.
- Validate design through simulations and analysis.

3. Functional Testing:

- Conduct initial tests to validate basic functionality, such as movement, steering, and control.

4. Software/Firmware Development:

- Develop and implement control algorithms, sensor integration, and user interface software.

5. Performance Optimization:

- Identify areas for improvement based on testing feedback.
- Optimize design, software, and components for better performance and reliability.

6. Final Prototype Testing:

- Conduct comprehensive testing of the RC fighting car prototype under various conditions.
- Ensure compliance with safety, quality, and regulatory requirements.

7. Project Closeout:

- Review project performance against initial goals and objectives.
- Document lessons learned and best practices for future projects.
- Celebrate project completion and team achievements.

Staffing Plan:

1. Project Manager:

- Oversee the entire project, including planning scheduling budgeting and resources allocation.
- Ensures that the projects stay on the track and meets its objectives within the specified timeframe and budget.

2. Software Engineering:

- Develops the firmware and software for the control the RC card, including Motor Control algorithms, sensor integrations and user interface design.
- Implements autonomous navigation and other advanced features.

3. Test Engineer:

- Develops test plans. Procedures and protocols for Validating the functionality and performance of the RC fighting car.
- Conducts both simulated and real-world testing to identify and resolve issues.

4. Manufacturing Engineer:

- Coordinates manufacturing process, including prototyping production and quality control.
- Ensures that the design specifications are translated into manufacturable products.

6. Customer Support Representative:

- Provides technical support troubleshooting assistance and product information to customers.
- Handles inquiries, complaints and feedback to ensure a positive customer experience.

Monitoring and Controlling Mechanism:

1. Control system:

- **Microcontroller or microprocessor:** The brain of the RC car responsible for possessing sensor data and generating control signals for the action actuators.

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- **Control algorithms:** Various console algorithms such as proportional-integral-derivative (PID) controller are used to stabilize the car maintaining desired trajectories and implement autonomous behaviours.
 - **Communication Interface:** Enables communication between the router controller and the card, often utilizing wireless protocols such as Bluetooth or Wi-Fi.

2. Actuators:

- **Motors:** Electric motors drive the wheels and other moving parts of the car. Speed controllers regulate the speed and direction of the motor based on control signals from the microcontroller.

3. User interface:

- **Remote controller:** Provides the user interface for sending commands to the car, including throttle, steering and mood selection.
- **Smartphone app:** Our RC car control by a dedicated app.

Risk Management:

1. Technical Risks:

- **Component failure:** Ayesha's physical components prone to failure, such as motors, batteries or electronic circuits. Unit debate decisions in five selecting high quality-quality components, conducting thorough testing and having backup system in place.
- **Interface and signal loss:** Wireless communication between the remote controller and the car may be susceptible to interference or signal loss. Implement robust communication protocols, use frequency-hopping spread spectrum (FHSS) techniques, and perform range testing to ensure reliable control.

2. Safety Risks:

- **Collision and Injury:** RC fighting cars may pose a risk of collision, particularly in multiplayer or combat scenarios. Implement safety features such as automatic emergency stop mechanisms, protective bumpers, and clear operating guidelines to minimize the risk of injury to users.
- **Electrical Hazards:** High-voltage components and exposed wiring can pose a risk of electric shock or fire. Ensure proper insulation, grounding, and circuit protection measures are in place to prevent electrical hazards.

3. Operational Risks:

- **User Error:** Inexperienced or careless users may mishandle the RC fighting car, leading to accidents or damage. Provide comprehensive user training, instructional materials, and safety guidelines to minimize the risk of user error.
- **Environmental Factors:** Adverse weather conditions, rough terrain, or obstacles in the operating environment can affect the performance and durability of the RC car.

Conduct field tests in various conditions to assess performance and identify potential hazards.

4. Legal and Regulatory Risks:

- **Compliance Issues:** Ensure that the RC fighting car complies with relevant safety standards, regulations, and licensing requirements. Conduct thorough risk assessments and consult legal experts to address any compliance issues.
- **Liability Concerns:** In the event of accidents or injuries involving the RC car, there may be legal liabilities and insurance implications. Obtain liability insurance coverage and implement liability waivers for users to mitigate legal risks.

5. Supply Chain Risks:

- **Supplier Delays:** Dependence on external suppliers for components or materials may lead to delays in production or cost overruns. Diversify the supply chain, maintain good relationships with suppliers, and have contingency plans for sourcing alternatives.

6. Security Risks:

- **Cybersecurity Threats:** Wireless communication and electronic control systems may be vulnerable to hacking or unauthorized access. Implement encryption, authentication, and access control measures to safeguard against cybersecurity threats.

List of Deliverables:

1. Conceptual Design:

- Concept sketches and renderings of the RC fighting car's appearance and features.
- Design specifications outlining the vehicle's size, weight, materials, and performance requirements.

2. Software/Firmware Development:

- Firmware code for controlling motors, sensors, and communication interfaces.
- User interface (UI) software for remote control, settings adjustment, and feedback display.

3. Testing and Validation:

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- Test plans and procedures for evaluating the performance, reliability, and safety of the RC car.
 - Test results documentation with observations, measurements, and analysis of test outcomes.

4. Quality Assurances:

- Quality control procedures and standards for ensuring consistency and reliability in manufacturing.
- Inspection reports and quality metrics monitoring the adherence to design specifications.

5. Deployment and Support:

- Deployment plan outlining the rollout strategy, training programs, and customer support channels.
- Warranty policy, service agreements, and maintenance schedules for ongoing support and updates.

Schedule Tracking Process:

1. Define Tasks and Milestones:

- Break down the project into smaller tasks and milestones, based on the project plan and timeline.
- Assign responsibilities to team members for each task or milestone.

2. Establish Baseline Schedule:

- Create a baseline schedule that outlines the planned start and end dates for each task and milestone.
- Set deadlines and target dates for key deliverables and project phases.

3. Monitor Progress Regularly:

- Regularly review the project schedule to monitor progress against the baseline.
- Update the schedule as tasks are completed or delayed, and adjust timelines accordingly.

4. Identify Delays and Issues:

- Identify any delays or issues that may impact the project schedule.
- Investigate the root causes of delays and take corrective actions to address them promptly.

5. Implement Risk Management Strategies:

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- Proactively identify and mitigate risks that could impact the project schedule.
 - Have contingency plans in place for handling unforeseen events or delays.

6. Review and Evaluate Performance:

- Conduct regular reviews of schedule performance to assess progress and identify areas for improvement.
- Evaluate the effectiveness of schedule tracking processes and make adjustments as needed.

Defect Tracking Process:

1. Identify the Defect:

- During operation, be observant of any malfunctions or unusual behaviour.
- This could include sluggish movement, control issues or damage after a fight.

2. Document the Defect:

- Note down the specific details of the defect.
- When did it occur? What actions were you taking? Are there any visual signs of damage (loose wires, broken parts)?
- Take pictures or videos if possible, for better reference.

3. Isolate the Problem:

- Try to narrow down the source by testing different components (try a different battery, check motor connections).
- Utilize resources like manuals, online forums, or consult with experienced RC enthusiasts to get insights.

4. Trace and Report:

- Use a tracking system (physical notebook, spreadsheet, dedicated software) to document each defect.
- Include details like date, description, troubleshooting steps taken, and any solutions attempted.
- This history will help identify recurring issues and improve future troubleshooting.

5. Resolution:

- Based on your findings, attempt repairs or replacements.
- Refer to manuals or online resources for repair guides specific to your RC fighting car model.

6. Verification and Improvement:

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- After repairs or replacements, thoroughly test the RC car to ensure the defect is resolved.
 - Update tracking system with the solution and outcome.

Conclusion:

The development of an RC fighting car involves careful planning, design, testing, and iteration to create a high-performance and engaging vehicle. By incorporating robust mechanical, electrical, and software components, along with effective defect tracking and schedule management processes, the project can successfully deliver a product that meets user expectations for performance, safety, and durability.